

NF Limiter

Complete User Manual - English



TRUE PEAK MASTERING LIMITER

Peak, loudness and stereo-image control with independent True Peak metering.

NF Audio Tools - By Neno Fernando

Interface overview

The interface is organized for immediate reading of input level, gain reduction, ceiling, loudness and true peak. Essential controls remain visible throughout operation.



1 Presets and comparison
Browser, A/B, COPY, SAVE and menu.

2 INPUT
Pre-processing L/R meters.

3 GAIN
Level sent into the limiting engine.

4 GAIN REDUCTION
Instant reduction and history.

5 Central metering
Peak, True Peak, LUFS-M, LUFS-I and OVER.

6 OUTPUT / CEILING
L/R output and maximum ceiling.

7 RELEASE / AUTO
Manual or adaptive recovery.

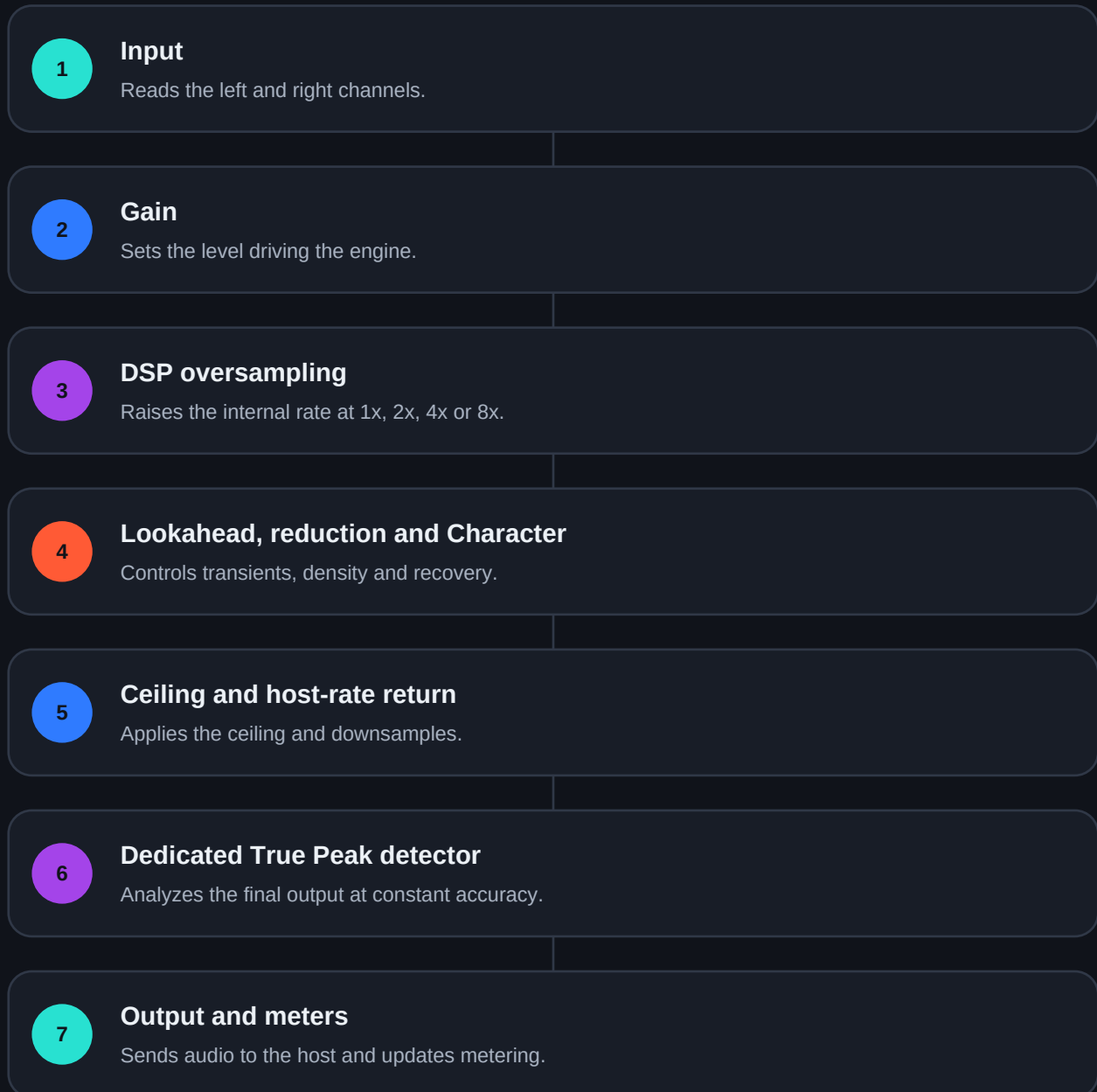
8 CHARACTER
CLEAN, PUNCH and LOUD.

9 TRUE PEAK / OS
Protection and 1x to 8x quality.

10 STEREO LINK / BYPASS
Stereo cohesion and comparison.

Internal signal flow

NF Limiter combines lookahead limiting, Character modes, selectable oversampling and an independent True Peak detector.



IMPORTANT

The selected oversampling factor controls processing quality. The True Peak detector is independent, so its accuracy does not change when switching 1x/2x/4x/8x.

Gain, input and reduction



INPUT and GAIN

INPUT meters show the incoming L/R levels. GAIN raises the level entering the limiter; more gain generally creates more reduction and loudness.

- Start with low GAIN and raise it while watching GAIN REDUCTION.
- Do not judge by loudness alone: compare at a similar perceived level.
- Different L/R peaks can reveal a stereo imbalance.

GAIN REDUCTION

The center display combines a numeric readout and history trace. More negative values mean deeper attenuation. Constant heavy reduction can soften transients and create excessive density. Use the history to see how the engine reacts over time.

Ceiling, output and safety



CEILING

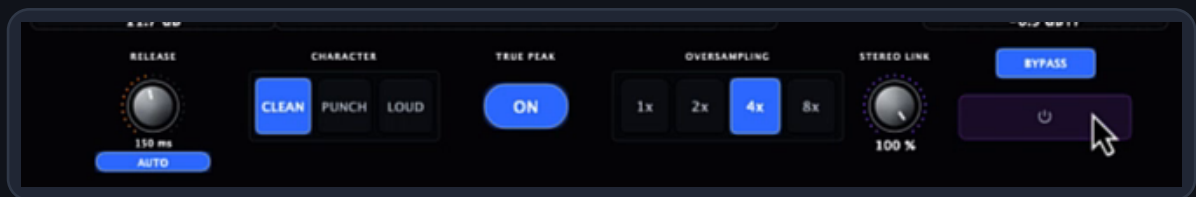
CEILING sets the output limit. With True Peak protection enabled, the engine accounts for reconstructed intersample peaks to reduce overs during conversion and encoding.

- -1.0 dBTP is a conservative starting point for digital delivery.
- Ceilings closer to 0 dBTP leave less margin.
- Ceiling choice does not replace distortion and loudness evaluation.

OUTPUT

OUTPUT meters show the channels sent to the host. Read them together with TRUE PEAK and TRUE PEAK OVER. The detector analyzes the final output after processing and downsampling.

Release and Auto mode



RELEASE

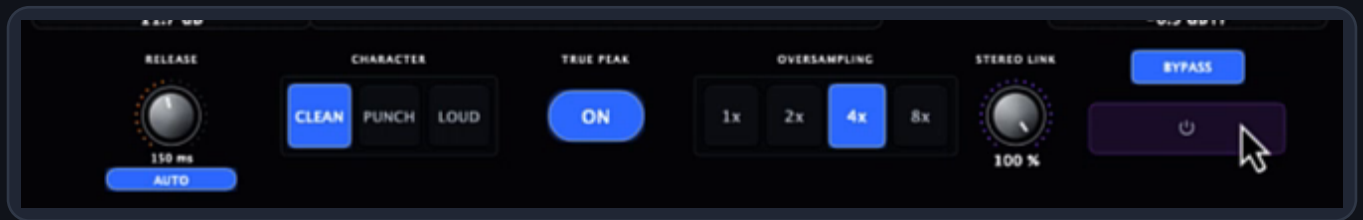
Release sets how quickly gain recovers after a peak is reduced. Short times can add energy but may sound rough or modulated. Long times are smoother but can hold the mix down.

- Short: more aggressive and energetic.
- Medium: balanced for general use.
- Long: smooth and stable recovery.

AUTO

AUTO adapts recovery to the material. It is the best starting point when a song alternates transients, sustain and changing density. Disable AUTO to set the release time manually in ms.

Character: CLEAN, PUNCH and LOUD



CLEAN

Prioritizes transparency, control and minimal coloration. Recommended for mastering, mix bus and already balanced material.

PUNCH

Favors impact and transient definition. Recommended for drums, rock, pop and mixes that must retain attack.

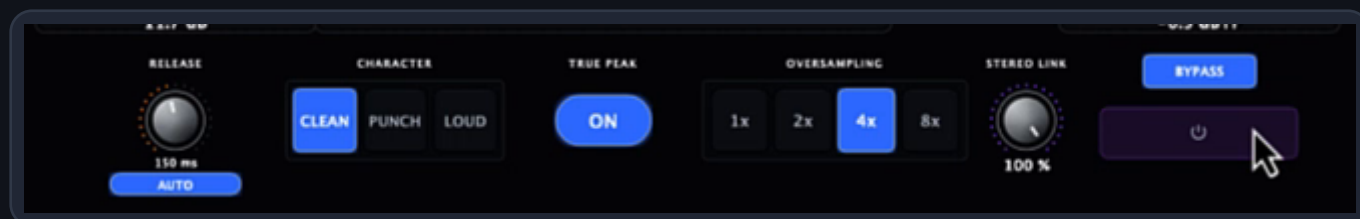
LOUD

Targets greater density and presence with a more intense nonlinear response. Watch saturation carefully.

Comparison tip

When changing Character, adjust GAIN to compare at a similar loudness. A louder mode can appear better simply because it is louder.

True Peak and Oversampling



TRUE PEAK

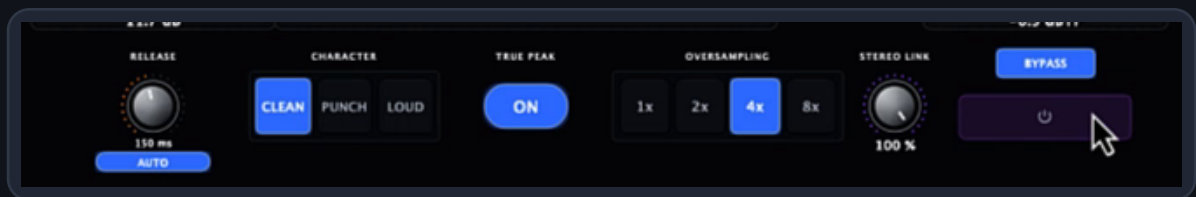
ON enables True Peak limiting. OFF disables only that protection; the True Peak meter and TRUE PEAK OVER warning continue monitoring the output. This lets you detect overs even when protection is off.

OVERSAMPLING

1x reduces CPU use. 2x and 4x balance quality and CPU. 8x provides the highest internal quality and lowest aliasing in nonlinear stages, at greater cost. The factor may affect latency and CPU use.

Factor	Suggested use	Tradeoff
1x	Editing and low CPU	Lowest internal quality
2x	Session and tracking	Good balance
4x	Mixing and mastering	High quality
8x	Final render/master	Highest CPU

Stereo Link and Bypass



STEREO LINK

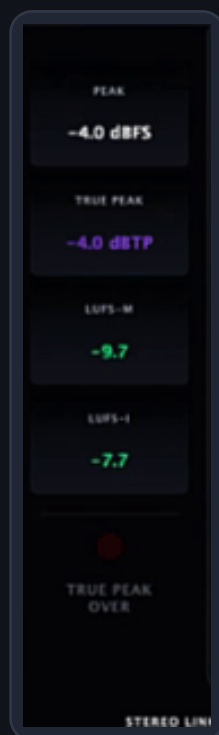
Controls how much the channels share gain reduction. At 100%, both sides react in a strongly linked way, preserving the position of elements in the panorama.

- 100%: the most stable and cohesive image.
- Middle values: more independence and movement.
- Low values: potential loudness increase, with risk of center shift.

BYPASS

Disables processing for comparison. Always compare at a similar perceived level. When bypassed, ceiling protection and limiting are inactive; monitor the level before switching.

Meters: Peak, True Peak and LUFS



Readout	What it represents	How to use it
PEAK / dBFS	Highest observed digital sample.	Technical sample-peak control.
TRUE PEAK / dBTP	Estimated reconstructed intersample peak.	Check margin for conversion and codecs.
LUFS-M	Momentary loudness over a short window.	Watch fast intensity changes.
LUFS-I	Integrated loudness since start/reset.	Evaluate the complete song or program.
TRUE PEAK OVER	Warning that the monitored limit was exceeded.	Review ceiling, gain and TP protection.

Responsible reading

LUFS alone does not define quality. Combine loudness, reduction, True Peak and critical listening. Platforms may apply normalization; there is no single ideal number for every genre or destination.

Presets, A/B and recall



Browser and SAVE

Use the arrows to browse presets and the center field to identify the current setting. **SAVE** opens the naming window and stores parameters as an internal preset. Use names that describe source, goal and intensity.

- Example: Master Clean -1 dBTP
- Example: Drum Punch 2 dB GR
- Avoid overwriting an important preset without a copy.

A/B and COPY

A and B hold two settings for fast comparison. **COPY** transfers the current state to the other slot; then change only the parameter you want to evaluate. This avoids relying on memory.

DAW recall

When a session is saved, the host should store NF Limiter parameters. On reopening, confirm **GAIN**, **CEILING**, **RELEASE**, **CHARACTER**, **TRUE PEAK**, **OVERSAMPLING** and **STEREO LINK**. Internal presets and session recall are related but different: a preset is reusable; recall belongs to the DAW project.

- Save the session after choosing a preset.
- Test recall before archiving an important project.
- A/B and temporary states do not replace a saved preset.

Recommended workflow

1

Insert it last

On a master, place the limiter after the main tonal processing.

2

Set the Ceiling

Choose destination headroom; -1.0 dBTP is a safe starting point.

3

Enable True Peak

Use protection to control reconstructed peaks.

4

Choose Oversampling

4x for balance; 8x for final render when CPU allows.

5

Choose Character

CLEAN for transparency, PUNCH for attack, LOUD for density.

6

Raise Gain

Watch reduction, loudness, True Peak and perceived quality.

7

Set Release and Link

Control breathing, stability and stereo image.

8

Compare and save

Use loudness-matched A/B and save a descriptive preset.

Starting points

The settings below are starting points. Adjust for the music, dynamics and final destination.

Application	Reduction	Ceiling	Character	Release	OS	Link
Transparent master	1-3 dB	-1.0 dBTP	CLEAN	AUTO	4x	100%
Dense master	3-6 dB	-1.0 dBTP	LOUD	AUTO	8x	80-100%
Pop/Rock	2-5 dB	-1.0 dBTP	PUNCH	AUTO/medium	4x-8x	80-100%
Drum bus	1-3 dB	-0.5 to -1	PUNCH	short/medium	2x-4x	60-100%
Vocal bus	1-3 dB	per chain	CLEAN/PUNCH	medium/AUTO	2x-4x	100%
Quick demo	1-4 dB	-1.0 dBTP	CLEAN	AUTO	2x	100%

Adjust by reduction, not fixed numbers

The same GAIN setting creates different reduction in every mix. Use GAIN REDUCTION as an operational reference and listening as the final decision. If bass causes pumping, try less Gain, a longer/Auto Release, or earlier mix treatment.

Troubleshooting and good practice

Symptom	Likely cause	Action
No reduction	Insufficient Gain or low signal.	Raise Gain and check INPUT.
Flattened sound	Excessive reduction.	Lower Gain or use CLEAN.
Pumping	Unsuitable Release or bass dominance.	Use AUTO/longer time; review the mix.
Harshness	Clipping/saturation or short Release.	Use less Gain, CLEAN or higher OS.
Center moves	Low Stereo Link.	Increase Stereo Link.
TRUE PEAK OVER	Final peak exceeded the monitored margin.	Lower Ceiling/Gain or enable TP.
High CPU	High OS and many instances.	Work at 2x/4x; render at 8x.
Preset not recalled	State was not saved/loaded.	Save preset and session; verify recall.

Hearing safety

Limiting raises average level. Monitor at a moderate volume, take breaks and lower monitoring before switching BYPASS on settings with high Gain.



Control	Essential function
GAIN	Drives the signal into the limiter.
CEILING	Sets the output limit.
RELEASE/AUTO	Controls recovery.
CHARACTER	Selects transparency, attack or density.
TRUE PEAK	Toggles protection; metering remains active.
OVERSAMPLING	Selects processing quality/cost.
STEREO LINK	Links channel reduction.
BYPASS	Compares processed and original.
A/B + COPY	Compares two settings.
SAVE	Stores an internal preset.

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